

Mutual Fund Analytics - Final Project Report

Abstract

This capstone project analyzes mutual fund data using Python, SQLite, SQL, Jupyter Notebook and Power BI.

Objectives

Build an ETL pipeline, clean data, compute performance metrics, perform EDA, create a recommendation engine and dashboard.

Technology Stack

Python, Pandas, NumPy, SQLite, SQL, Power BI, Matplotlib, Jupyter Notebook.

Datasets

Fund master, NAV history, AUM, SIP inflows, category inflows, folio counts, scheme performance, investor transactions, portfolio holdings and benchmark indices.

ETL Pipeline

Data ingestion, cleaning and metric computation are orchestrated using scripts/etl_pipeline.py.

Database

SQLite database stores fact_nav, fact_transactions and fact_performance tables.

Analytics

EDA and advanced analytics identify trends, risk metrics and fund performance.

Performance Metrics

Average returns, alpha, beta, Sharpe ratio, Sortino ratio, standard deviation, maximum drawdown, expense ratio and AUM are computed.

Recommendation Engine

A Python-based recommendation module suggests funds based on available performance information.

Dashboard

Power BI dashboard visualizes fund performance, NAV trends, investor activity and portfolio insights.

Conclusion

The project demonstrates a complete analytics workflow from data ingestion to visualization and reporting.